A Low-z / Mid-z Nuisance Mode in the Union3 Compressed Supernova Data: Implications for the DESI DR1 Dynamic Dark Energy Preference

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Abstract

We test whether the Union3 supernova contribution to the DESI DR1 CPL dynamical dark-energy preference is robust to a single redshift-dependent nuisance direction. Using the public Union3 compressed 22-node precision matrix and individual light-curve fit files, we find that a Gaussian low-z / mid-z nuisance template $T(z) = G(z; 0.09, 0.06) - G(z; 0.775, 0.10)$ improves the fixed-$\Omega_m = 0.2975$ fit by $\Delta\chi^2 = 17.12$ (AIC +15.12), compared with $\Delta\chi^2 \approx 7.53$ (AIC $\approx +3.53$) for CPL. After including this nuisance, CPL adds only $\Delta\chi^2 \approx 0.87$ and is AIC-disfavored. A public-data Cobaya cross-check using DESI DR1 BAO, a compressed CMB prior on $\Omega_m h^2$, and the Union3 compressed likelihood gives the same qualitative result: CPL improves χ^2 by 8.29 without the nuisance mode, but only by 0.52 after the nuisance mode is included, with the CPL+nuisance best-fit solution shifting close to ΛCDM ($w_0 = -0.999$, $w_a = -0.192$). **A new full-likelihood cross-check**—Planck NPIPE CamSpec TTTEEE plus low-ℓ TT/EE, DESI DR1 BAO, and the Union3 likelihood with the nuisance amplitude sampled directly (four MCMC chains; the means converged to $R - 1 = 0.047$, the bounds criterion more weakly, $R - 1 \approx 0.15$)—confirms this: without the nuisance, CPL is preferred over ΛCDM by AIC gain +8.42 ($w_0 = -0.677 \pm 0.113$, $w_a = -1.136 \pm 0.438$); once the nuisance is marginalized the CPL preference is removed (AIC gain −0.38 relative to ΛCDM+nuisance), with the posterior shifting to $w_0 = -0.933 \pm 0.129$, $w_a = -0.501 \pm 0.400$ and a recovered amplitude $a = 0.029 \pm 0.010$ consistent with the compressed-data value. Additional robustness tests (81 template variants; a 200-template random null test; leave-one-region-out; cumulative low-z node removal) strengthen the diagnostic. The dynamical-dark-energy preference is weakened from $\sim 2.4\sigma$ to $\sim 1.6\sigma$ but not entirely eliminated, consistent with the partial weakening already seen in the official Union3.1 joint analyses. The definitive test remains to rerun the official DESI+CMB+Union3 w_0–w_a likelihood with an added redshift-dependent nuisance parameter.

1 Introduction

The DESI 2024 Data Release 1 BAO analysis, combined with CMB and Type Ia supernova data, reports a preference for dynamical dark energy described by the CPL equation of state $w(a) =$

$w_0 + w_a(1-a)$, with a combined significance of approximately 3.4–3.5σ for DESI+CMB+Union3 [1]. The Union3 compilation [2] contains 2,087 Type Ia supernovae from 24 surveys spanning approximately 7 billion years of cosmic history, compressed into a 22-node distance-modulus spline with an associated precision matrix. Individual SN surveys cover limited redshift ranges, making multi-survey compilations necessary but potentially introducing redshift-dependent systematics at survey boundaries. Related work by Efstathiou [5] identified a systematic offset between low- and high-redshift supernovae in Pantheon+ and DES5Y, and noted that Union3 individual-SN data were not publicly available for direct analysis. Subsequent analyses have sharpened this question: Huang et al. [11] used the intercept of the SN magnitude–redshift relation to argue that correcting a low-z discrepancy in DES5Y reduces the CPL preference to 0.5–1.5σ, and Ormondroyd et al. [12] showed via Bayesian model selection that a low-z magnitude offset is favoured over a dynamical reconstruction for DES5Y+DESI; most recently Mirpoorian et al. [13] fit a look-back-time-scaled magnitude nuisance jointly with w_0w_aCDM on Pantheon+, finding negligible impact for uncalibrated SNe. These interpretations remain contested, with Vincenzi et al. [14] attributing much of the offset to Malmquist-bias and intrinsic-scatter modelling and the recalibrated DES "Dovekie" analysis [15] continuing to prefer evolving dark energy. Almost all of this work targets DES5Y or Pantheon+, whose per-object data are public, which is what makes the Union3-specific test below distinct. Cortês & Liddle [8], addressing DESI DR2, further showed that properly accounting for the substantial overlap between primary SN compilations reduces the combined ΛCDM exclusion significance to 3.1σ—identical to the DESI BAO+CMB result without any supernova data; their methodological argument about compilation overlap is relevant to DR1-style compilation comparisons as well. Independent forecasting work by Truong et al. [10], evaluated at the same DESI+CMB+Union3 best fit, shows that the w_0–w_a discriminating power of SNe Ia is concentrated at $z \lesssim 0.6$, with low-redshift supernovae primarily constraining w_0 and a medium-redshift anchor required to make them informative; a coherent low-z/mid-z distance distortion is therefore precisely the kind of systematic direction that can mimic or absorb the dynamical-dark-energy signal, which motivates the template form adopted here. The present work performs, to the best of the author's knowledge, the first compressed-data robustness test of this kind specifically for Union3, using the publicly available `lcfit_Union3.tar.gz` and compressed precision matrix from the `rubind/union3_release` repository.

2 Data and Method

We use the public files `mu_mat_union3_cosmo=2_mu.fits` and `lcfit_Union3.tar.gz` from the `rubind/union3_release` GitHub repository [9]. The FITS matrix structure is: `mat[0,1:]` $= z$ nodes (22 values), `mat[1:,0]` $= \mu$ values, `mat[1:,1:]` $=$ precision matrix P (22×22). The magnitude offset M is analytically marginalized in all fits via GLS; the nuisance amplitude a is also analytically marginalized when included. For node-removal tests, the covariance matrix $C = P^{-1}$ is subselected and re-inverted correctly. The primary nuisance template is

$$T(z) = G(z; 0.09, 0.06) - G(z; 0.775, 0.10), \tag{1}$$

standardized to zero mean and unit RMS, where $G(z; c, s) = \exp[-\frac{1}{2}((z-c)/s)^2]$. Template centers were placed at known survey-composition transitions (not optimized over the data): the low-z anchor region dominated by SDSS/Foundation/Pan-STARRS ($z \approx 0.09$) and the mid-z transition into SNLS/ESSENCE/DES3_Deep ($z \approx 0.775$). Quality-cut pass fractions differ between these zones: 0.718 at $z \approx 0.10$ versus 0.952 at $z \approx 0.75$.

For the public joint cross-check (Table 2) we use Cobaya with CAMB, combining the official DESI DR1 BAO likelihood (`bao.desi_2024_bao_all`, with its own covariance), a compressed CMB prior on $\Omega_m h^2 = \Omega_b h^2 + \Omega_c h^2 + \Omega_\nu h^2 = 0.1430 \pm 0.0020$ (Planck 2018 [6]; deliberately broadened from the Planck best-fit $\sigma = 0.0011$ as a conservative choice), and the Union3 compressed likelihood with the nuisance template implemented as a custom Cobaya likelihood. Sampled parameters: H_0 (flat 55–85 km/s/Mpc), $\Omega_b h^2$ (Gaussian 0.02237 ± 0.00015), $\Omega_c h^2$ (flat 0.09–0.15), w_0 (flat -2 to -0.3), w_a (flat -3 to 2), the SN magnitude offset M (flat -1 to 1 mag) and, when included, the nuisance amplitude a (flat -0.08 to 0.08 mag); $\tau = 0.054$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.965$ and $\Sigma m_\nu = 0.06$ eV are fixed, flat geometry is assumed, and the sound horizon is computed by CAMB. In these joint runs M and a are sampled rather than analytically marginalized (unlike the compressed-data fits above); the exact configurations and the custom likelihood are provided in the repository (`cobaya/joint_public_*.yaml`, `cobaya/union3_nuisance_likelihood.py`, `cobaya/cmb_omh2_prior.py`). We note that Kim [7] showed that the Union3 compressed product distributes a posterior under a spline-node prior rather than a raw likelihood, which explains why proxy baseline CPL best-fits differ slightly from the official DESI result; this does not affect the relative CPL vs CPL+nuisance comparison.

3 Union3 Compressed-Data Results

Table 1 summarises the protocol verification. The core fixed-Ω_m diagnostics reproduce to rounding precision. A low-minus-mid step template also gives a strong improvement, confirming that the effect is not specific to the Gaussian functional form.

Table 1: Protocol verification summary.

Test	Claimed	Verified	Status
Ω_m best-fit (ΛCDM free)	0.356	0.3559	Reproduced
χ^2 at best-fit	23.96	23.958	Reproduced
DESI-like Ω_m penalty	5.15	5.152	Reproduced
CPL $\Delta\chi^2$ / AIC	7.53 / +3.53	7.536 / +3.536	Reproduced
Low-mid Gaussian $\Delta\chi^2$ / AIC	17.12 / +15.12	17.120 / +15.120	Reproduced
Color $\Delta\chi^2$ / AIC	6.52 / +4.52	2.947 / +0.947	Weighting-dependent
CPL after low-mid / AIC	0.87 / -3.13	0.870 / -3.130	Reproduced
Low-mid prior threshold $\sigma_a \geq 0.005$ mag	0.005 mag	RMS	Reproduced

4 Joint DESI DR1 BAO + CMB + Union3 Public Cross-Check (compressed CMB prior)

We ran a public-data Cobaya cross-check using the DESI DR1 BAO likelihood, a compressed CMB prior, and the Union3 compressed likelihood. This is stronger than a simple proxy grid scan, but it is still not the official DESI+CMB+Union3 full likelihood because the CMB information is compressed and the Union3 nuisance parameter is added externally.

Without the nuisance mode, CPL improves χ^2 by 8.29 and shifts to $w_0 = -0.688$, $w_a = -1.070$, consistent with published DESI DR1+CMB+Union3 results. After including the nuisance, CPL improves χ^2 by only 0.52 relative to ΛCDM+nuisance (AIC -3.48), while the CPL+nuisance best-fit solution shifts to $w_0 = -0.999$, $w_a = -0.192$—close to ΛCDM in this proxy analysis.

Table 2: Public-data joint model comparison (compressed CMB prior). AIC gain = $\Delta\chi^2 - 2k$, where k is the number of additional parameters vs ΛCDM. The highlighted row shows that CPL adds only $\Delta\chi^2 = 0.52$ once the nuisance direction is included.

Model	H_0	w_0	w_a	a_{nuis}	best-sample χ^2	$\Delta\chi^2$ / AIC gain
ΛCDM	68.77	—	—	—	41.24	—
CPL	66.64	−0.688	−1.070	—	32.95	+8.29 / +4.29
ΛCDM + nuis.	69.09	—	—	0.0279	25.04	+16.20 / +14.20
CPL + nuisance	69.79	−0.999	−0.192	0.0306	24.51	+16.73 / +10.73
CPL vs ΛCDM+nuis.						+0.52 / −3.48

5 Full-Likelihood Cross-Check (Planck NPIPE CamSpec + DESI DR1 BAO + Union3)

We repeated the four-model comparison with the full likelihood, replacing the compressed CMB prior of Section 4 with Planck NPIPE CamSpec TTTEEE plus low-ℓ TT/EE, and adding the redshift-dependent nuisance parameter directly inside the Union3 likelihood (amplitude a). Four MCMC chains were run per model; the Gelman–Rubin convergence on the parameter means reached $R-1 = 0.047$, while the more stringent bounds (credible-interval) criterion was weaker, $R-1 \approx 0.15$. The full-likelihood comparison should therefore be regarded as a public cross-check rather than collaboration-grade parameter estimation. This realizes the falsification test of Section 10 within the public pipeline, with the nuisance mode marginalized rather than added externally.

Table 3: Full-likelihood model comparison (Planck NPIPE CamSpec TTTEEE + low-ℓ TT/EE + DESI DR1 BAO + Union3). Four MCMC chains per model (means $R-1 = 0.047$, bounds $R-1 \approx 0.15$). Quoted χ^2 are the best values among the MCMC samples (no separate minimizer was run); the present table is therefore an MCMC best-sample cross-check, and a separate minimizer run would be required for final best-fit publication numbers. AIC gain = $\Delta\chi^2 - 2k$ relative to ΛCDM; positive = favoured. The highlighted comparison shows that once the nuisance direction is marginalized, the CPL parameters (w_0, w_a) no longer improve AIC.

Model	H_0	w_0	w_a	a_{nuis}	best-sample χ^2	$\Delta\chi^2$ / AIC gain
ΛCDM	67.8	—	—	—	11007.56	—
CPL	66.4	-0.677 ± 0.113	-1.136 ± 0.438	—	10995.14	+12.42 / +8.42
ΛCDM + nuis.	68.0	—	—	0.027 ± 0.010	10992.23	+15.33 / +13.33
CPL + nuisance	69.3	-0.933 ± 0.129	-0.501 ± 0.400	0.029 ± 0.010	10988.61	+18.95 / +12.95
CPL + nuisance *vs* ΛCDM + nuisance						+3.62 / **−0.38**

Without the nuisance mode, CPL is strongly preferred over ΛCDM ($\Delta\chi^2 = 12.42$, AIC gain +8.42), with $w_0 = -0.677 \pm 0.113$, $w_a = -1.136 \pm 0.438$—a posterior mean lying 2.4σ from ΛCDM in the w_0–w_a plane (Gaussian-equivalent of the 2D Mahalanobis distance $r = 2.86$),[1] consistent with the published DESI DR1+CMB+SN preference. Including the nuisance parameter yields $a = 0.029 \pm 0.010$ ($\sim 2.8\sigma$ from zero, consistent with the compressed-data amplitude) and shifts the posterior to $w_0 = -0.933 \pm 0.129$, $w_a = -0.501 \pm 0.400$, now 1.6σ from ΛCDM ($r = 2.11$).

[1] The equivalent one-dimensional significance is the two-sided Gaussian tail enclosing the same probability as the 2D Mahalanobis ellipse, $P = 1 - \exp(-r^2/2)$; i.e. $\mathrm{erf}(\sigma/\sqrt{2}) = 1 - \exp(-r^2/2)$. This maps $r = 2.86 \to 2.4\sigma$ and $r = 2.11 \to 1.6\sigma$.

The decisive comparison is CPL+nuisance versus ΛCDM+nuisance. Once the nuisance direction is marginalized, adding the CPL parameters improves the best-fit χ^2 by only 3.62 for two extra parameters, corresponding to an AIC gain of -0.38: the strong AIC preference for dynamical dark energy (+8.42 relative to ΛCDM without the nuisance) is removed. The full-likelihood result therefore confirms the compressed-data analysis: the Union3 low-z/mid-z nuisance direction is favoured by the data and absorbs most of the dynamical-dark-energy preference. As in the compressed proxy, the effect weakens but does not entirely eliminate the preference (the shift is from $\sim 2.4\sigma$ to $\sim 1.6\sigma$, and the residual CPL AIC penalty is marginal), consistent with the partial weakening already seen in the official Union3.1 joint analyses (Section 6).

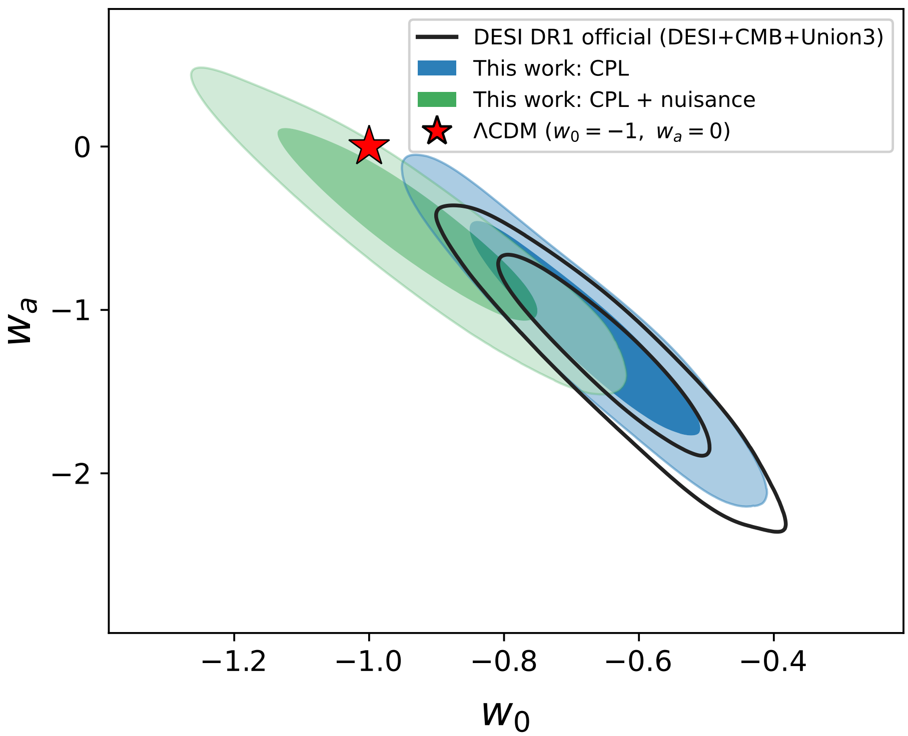

Figure 1: Full-likelihood joint cross-check in the w_0–w_a plane. Black contours: DESI DR1 official w_0w_aCDM posterior (DESI+CMB+Union3, `plik` TT,TE,EE). Blue: this work's CPL (NPIPE CamSpec); green: this work's CPL+nuisance. Red star marks ΛCDM ($w_0 = -1$, $w_a = 0$). This work's CPL reproduces the official DESI contour ($w_0 = -0.68$, $w_a = -1.14$ vs DESI $w_0 = -0.64$, $w_a = -1.30$), the small offset being consistent with the different CMB likelihood choice; marginalizing the Union3 nuisance direction (green) shifts the posterior toward ΛCDM ($w_0 = -0.93$, $w_a = -0.50$), from $\sim 2.4\sigma$ to $\sim 1.6\sigma$.

6 Three-Version Progression: Union3 to Union3.1

Hoyt et al. [3] published host-galaxy mass re-measurements for approximately 2,000 Union3 supernovae, finding that low-z (< 0.10) host-galaxy stellar masses were systematically overestimated, reducing a previously reported > 0.03 mag low-z distance offset to ≈ 0.01 mag. The updated cosmological framework (UNITY1.8) was presented by Rubin et al. [4]. Applying the same nuisance template to all three public compressed matrices shows the nuisance amplitude decreasing monotonically, while the CPL AIC gain becomes small by AIC comparison in the Union3.1 versions. The nuisance amplitude decrease of 0.0056 mag (0.0279 → 0.0223) is of the same order of magnitude as the relative low-z distance shift of 0.010–0.013 mag applied in the Union3.1 corrections. This correspondence was not built into the template—the amplitude emerged from fitting the compressed data, and is consistent with the template tracing a low-z systematic direction that is reduced, but not removed, by the Union3.1/UNITY processing. This monotonic decrease also bears directly on the central interpretive question of whether the template absorbs a genuine systematic or, instead, real cosmological signal: the underlying w_0–w_a cosmology is identical across the three compressed matrices, so a template tracing a purely cosmological signal would not naturally be expected to decrease in step with documented low-z host-mass corrections. Instead the amplitude falls in step with the documented low-z host-mass correction of Hoyt et al. [3]—the behaviour expected if the template is tracing that corrected systematic rather than the dynamical-dark-energy signal. Importantly, this comparison is a heuristic compressed-data diagnostic only and should not be conflated with the updated official joint analyses: Hoyt et al. [3] report a $\approx 3.4\sigma$ deviation from ΛCDM in the official BAO+CMB+Union3.1 combination, and Rubin et al. [4] report $\approx 2.6\sigma$ tension with UNITY1.8. The official joint preference is thus weakened but not eliminated by the Union3.1 corrections.

Table 4: Three-version comparison. [a] CPL $\Delta\chi^2$ values marked ≈ are grid-scan approximations with ≈ 0.02 uncertainty; the scipy-optimize value for Union3 original is 7.536 (Table 1). Nuisance amplitude decreases monotonically across all three versions.

Metric	Union3 original	Union3.1 UNITY1.7	Union3.1 UNITY1.8
Ω_m best-fit	0.3559	0.3424	0.3340
DESI penalty $\Delta\chi^2$	5.152	3.256	2.347
CPL $\Delta\chi^2$ [a]	≈ 7.52	≈ 4.79	≈ 4.74
CPL AIC gain [a]	≈ +3.52	≈ +0.79	≈ +0.74
Low-mid Gaussian $\Delta\chi^2$	17.120	12.474	11.958
Low-mid Gaussian AIC gain	+15.12	+10.47	+9.96
Nuisance amplitude (mag)	0.0279	0.0233	0.0223
CPL after nuisance AIC	−3.13	−2.80	−2.04
Low-z shift vs Union3	—	−0.010	−0.013

7 Additional Robustness Tests

The robustness tests in this section use the same fixed-$\Omega_m = 0.2975$ protocol as Table 1; their $\Delta\chi^2$ values are directly comparable to the Table 1 results and are reproducible from the GitHub repository CSV outputs. CPL+nuisance $\Delta\chi^2$ values in reduced-node fits can be optimizer-sensitive and should not be interpreted as precise parameter estimates.

7.1 Template Robustness Grid

To test whether the result depends on the exact Gaussian centers and widths, we scanned 81 nearby low-z / mid-z template variants. The nuisance improvement remains positive in all 81 cases. The conclusion does not depend on the primary template sitting near the top of this grid (original 17.12 vs grid maximum 17.45): even the grid *median* (13.18) comfortably exceeds the CPL improvement (7.54), so any representative low-z/mid-z template in this family out-performs CPL.

Table 5: Template robustness grid summary.

Quantity	Value
Number of templates	81
ΛCDM χ^2 (fixed Ω_m)	29.110
CPL χ^2 (fixed Ω_m)	21.574
CPL $\Delta\chi^2$ vs ΛCDM	7.536
Original nuisance $\Delta\chi^2$	17.120
Grid median nuisance $\Delta\chi^2$	13.176
Grid min nuisance $\Delta\chi^2$	9.666
Grid max nuisance $\Delta\chi^2$	17.454
Median CPL $\Delta\chi^2$ after nuisance	0.413

7.2 Random Smooth-Template Null Test

A random smooth-template null test compares the original low-z / mid-z template with 200 random smooth redshift templates generated as follows: each random template is a difference of two Gaussians with centers drawn uniformly from $z \in [0.05, 0.40]$ and $z \in [0.40, 1.50]$ respectively, and widths drawn uniformly from $[0.04, 0.20]$, with random sign; each template is then standardized to zero mean and unit RMS on the 22 nodes before fitting (random seed 42 for reproducibility). No random template reached the original template improvement; conservatively, the empirical tail probability is $p \approx 1/201$ (add-one estimator).

Table 6: Random smooth-template null test (fixed-$\Omega_m = 0.2975$ protocol). The original template $\Delta\chi^2 = 17.120$; the maximum random template reached only 12.433. The 0/200 exceedance result is a conservative ranking diagnostic rather than proof of template uniqueness.

Quantity	Value
Number of random templates	200
Original template $\Delta\chi^2$ (fixed Ω_m)	17.120
Random median $\Delta\chi^2$	1.357
Random 95th percentile $\Delta\chi^2$	8.997
Random max $\Delta\chi^2$	12.433
Empirical tail probability	0/200; add-one $p \approx 1/201$ (≈ 0.005)

A sharper version of this test isolates the structure the template captures *beyond* CPL. For each template we compute the improvement it adds once CPL is already in the model, $\Delta\chi^2_{\text{after CPL}} = \chi^2(\text{CPL}) - \chi^2(\text{CPL} + \text{template})$, which removes the part of the residual that CPL can itself absorb (cf. Section 7.6). Applied to the same 200-template ensemble, the original low-z/mid-z template adds $\Delta\chi^2 = 10.45$ after CPL, exceeding *every* random smooth template (maximum 8.99, median 1.33; add-one $p \approx 1/201$; Table 7). The structure the template captures beyond CPL is therefore not generic overfitting of a flexible direction: no random smooth template orthogonal-enough to CPL reaches it. As in the standalone test this is a conservative ranking diagnostic rather than a proof of uniqueness.

Table 7: Beyond-CPL (orthogonalized) null test: improvement added *after* CPL, $\Delta\chi^2_{\text{after CPL}}$, for the original template versus 200 random smooth templates (same recipe and fixed-$\Omega_m = 0.2975$ protocol as Table 6). Removing the CPL-absorbable part lowers the random ceiling from 12.4 to 9.0, yet the original template still exceeds all 200.

Quantity	Value
Original template $\Delta\chi^2_{\text{after CPL}}$	10.45
Random median $\Delta\chi^2_{\text{after CPL}}$	1.33
Random 95th percentile $\Delta\chi^2_{\text{after CPL}}$	5.42
Random max $\Delta\chi^2_{\text{after CPL}}$	8.99
Empirical tail probability	0/200; add-one $p \approx 1/201$ (≈ 0.005)

7.3 Leave-One-Redshift-Region-Out Test

We removed broad redshift intervals one at a time and repeated the four-model compressed-data fit. The nuisance improvement remains substantial in every cut, while CPL adds little after nuisance is included. The largest weakening occurs when $z = 0.10$–0.30 is removed, localizing part of the signal to that low-redshift transition region.

Table 8: Leave-one-redshift-region-out robustness test. Only $\Delta\chi^2$ differences are interpreted; cosmological parameter values in reduced-node fits are diagnostic and may be weakly constrained.

Case	Removed z	N kept	$\Delta\chi^2$ CPL	$\Delta\chi^2$ nuisance	CPL after nuisance
Full sample	none	22	7.54	17.12	0.87
Very low z	0.00–0.10	21	7.58	16.76	0.82
Low z	0.10–0.30	18	2.80	7.65	0.48
Mid z	0.30–0.70	14	13.26	17.57	0.30
High-mid z	0.70–1.00	17	7.19	12.10	0.46
High z	1.00+	18	8.05	17.51	1.38

7.4 Cumulative Low-z Node Removal

As a complement to the broad leave-one-region-out test, we progressively removed low-redshift nodes below successively higher thresholds and re-evaluated CPL on the remaining nodes. This directly tests the concern that the nuisance template may effectively remove low-z supernova information. Removing only the single lowest node ($z < 0.08$ or < 0.10) leaves CPL essentially unchanged ($\Delta\chi^2 \approx 7.58$). The CPL preference falls sharply only when nodes up to $z < 0.12$ are removed ($\Delta\chi^2 = 2.29$, AIC -1.71) and is nearly eliminated by $z < 0.20$ ($\Delta\chi^2 = 0.87$, AIC -3.13). This localizes the signal to the $z = 0.10$–0.20 transition region, consistent with the low-z / mid-z nuisance template and not attributable to a single outlier node.

7.5 Low-z / Mid-z Component Decomposition

To see which redshift zone drives the result, and to address the concern that the low-z arm overlaps the peculiar-velocity region, we fit the two Gaussian arms of T as independent nuisance amplitudes. Individually, the low-z arm $G(z; 0.09, 0.06)$ gives $\Delta\chi^2 = 11.69$ and the mid-z arm $G(z; 0.775, 0.10)$

Table 9: CPL preference after cumulative removal of low-redshift nodes. The sharp drop between $z < 0.10$ and $z < 0.12$ localizes the signal to the $z = 0.10$–0.20 survey-transition region.

Removed nodes	N kept	CPL $\Delta\chi^2$	CPL AIC gain
None (full sample)	22	7.536	+3.536
$z < 0.08$	21	7.581	+3.581
$z < 0.10$	21	7.581	+3.581
$z < 0.12$	20	2.286	−1.714
$z < 0.20$	19	0.871	−3.129
$z < 0.30$	17	1.155	−2.845

gives $\Delta\chi^2 = 4.81$; fitted together (two amplitudes) they give $\Delta\chi^2 = 17.13$, essentially the single-template value, confirming the two arms are nearly independent in the precision metric. The CPL-absorbing power is concentrated in the low-z arm: CPL adds only $\Delta\chi^2 = 0.88$ after the low-z arm alone, but still $\Delta\chi^2 = 7.45$ after the mid-z arm alone—the mid-z arm is nearly orthogonal to CPL. Both arms nonetheless carry beyond-CPL structure (5.04 and 4.73 added after CPL, respectively).

Crucially, the cumulative node-removal test (Table 9) localizes the low-z power to $z \approx 0.10$–0.20, not to the lowest node $z = 0.05$ where peculiar velocities dominate: removing nodes only up to $z < 0.10$ leaves CPL essentially unchanged, and the preference collapses only once $z < 0.12$–0.20 is removed. The low-z arm is thus tracing the $z \approx 0.10$–0.20 survey-transition region—the same low-z range corrected by the Hoyt et al. [3] host-mass re-measurement (Section 6)—rather than the very-low-z peculiar-velocity zone.

Table 10: Decomposition of the template into its two Gaussian arms (fixed $\Omega_m = 0.2975$ protocol). The low-z arm carries most of the CPL-absorbing power; the mid-z arm is nearly orthogonal to CPL.

Component	$\Delta\chi^2$ alone	CPL after it	it after CPL
Low-z arm $G(z; 0.09, 0.06)$	11.69	0.88	5.04
Mid-z arm $G(z; 0.775, 0.10)$	4.81	7.45	4.73
Both arms (2 amplitudes)	17.13	–	–
Full template T	17.12	0.87	10.45

7.6 Template–CPL Overlap (Degeneracy Test)

A natural concern is that the template $T(z)$ merely reproduces the CPL distance deformation, so that its improvement is the dynamical–dark-energy signal relabeled. We test this directly in the Union3 precision metric, with the marginalized offset M projected out. The angle between T and the CPL best-fit deformation $\delta\mu_{\rm CPL} = \mu_{\rm CPL} - \mu_{\Lambda{\rm CDM}}$ is $\cos\theta = 0.79$ ($\theta \approx 38^\circ$); projecting T onto the full CPL tangent subspace $\{\partial\mu/\partial w_0,\ \partial\mu/\partial w_a\}$ gives the same value, with 63% of T lying inside the CPL subspace and 37% orthogonal to it. The two directions are thus strongly aligned but not degenerate. The nested $\Delta\chi^2$ decomposition (Table 11) is markedly asymmetric: once T is included CPL adds only $\Delta\chi^2 = 0.87$, whereas once CPL is included T still adds $\Delta\chi^2 = 10.45$. CPL is therefore almost entirely contained in T (88% of its pull is shared), while most of T's improvement is structure CPL cannot produce, localized at the $z \approx 0.78$ survey transition (Figure 2). Consistently, the recovered amplitude *persists* when CPL is free in the model: $a = 0.0279$ for ΛCDM+nuisance (matching the compressed-data value) and $a = 0.0357$ for CPL+nuisance.

Were T merely the dark-energy signal in disguise, θ would vanish, T would add nothing after CPL, and a would collapse toward zero; none of these holds. This rules out exact T–CPL degeneracy. It does not by itself distinguish a systematic from a signal—goodness-of-fit cannot—which is why the discriminating evidence remains the host-mass correspondence of Section 6 and the survey-composition transitions, not the fit alone. The large $\Delta\chi^2 = 10.45$ that T adds after CPL reflects genuine residual structure rather than overfitting. The beyond-CPL null test of Section 7.2 provides the direct overfitting check for this degeneracy concern: after CPL is already fitted, the original template still adds $\Delta\chi^2 = 10.45$, exceeding every random smooth template in the same after-CPL test (maximum 8.99; add-one $p \approx 1/201$). The residual structure T captures is therefore not typical of a random smooth direction once the CPL-absorbable component has been removed. The quantity relevant to the dark-energy claim is in any case the 0.87 that CPL adds after T, not the 10.45.

Table 11: Template–CPL overlap (fixed $\Omega_m = 0.2975$ protocol). The angle θ is measured in the Union3 precision metric with the magnitude offset M marginalized. All inputs reproduce the Table 1 diagnostics to rounding precision.

Quantity	Value
$\cos\theta$ (T vs CPL deformation)	0.79
θ (T vs CPL 2D subspace)	37.6°
T inside CPL subspace ($\cos^2\theta$)	63%
T orthogonal to CPL ($\sin^2\theta$)	37%
CPL $\Delta\chi^2$ added after T	0.87
T $\Delta\chi^2$ added after CPL	10.45
shared (overlap) $\Delta\chi^2$	6.67
amplitude a, ΛCDM+nuisance (mag)	0.0279
amplitude a, CPL+nuisance (mag)	0.0357

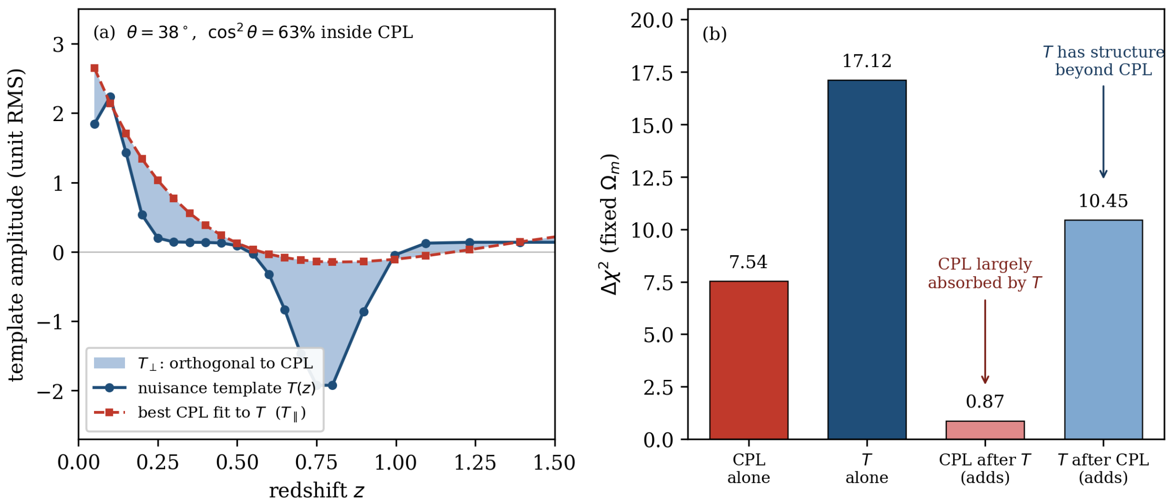

Figure 2: Template–CPL overlap in the Union3 precision metric. **(a)** The nuisance template $T(z)$ (blue) and its best projection onto the CPL subspace, $T_\parallel$ (red dashed). CPL reproduces T at low redshift but cannot fit the sharp $z \approx 0.78$ step (shaded $T_\perp$, the 37% of T orthogonal to all w_0–w_a deformations). **(b)** Nested $\Delta\chi^2$ ledger: CPL adds only 0.87 after T, but T adds 10.45 after CPL: CPL is largely absorbed by T, while T contains structure beyond CPL.

8 Independent Reproduction

An independent reproduction of the Union3.1 UNITY1.8 results directly from the raw FITS matrix (without astropy, using manual header parsing and big-endian float64 reading) confirmed: $\Delta\chi^2(\text{CPL}) = 4.7393$, $\Delta\chi^2(\text{Gaussian template}) = 11.9581$, $\Delta\chi^2(\text{low-minus-mid step}) = 11.7335$, and $\Delta\chi^2(\text{CPL after step}) = 0.1243$. All values match to rounding precision. The reproduction also confirmed that node-removal tests must subselect the covariance matrix $C = P^{-1}$ and re-invert; subsetting the precision matrix directly gives incorrect results.

9 Limitations

- Compressed-data and public-data proxy analyses, plus a full-likelihood cross-check in which the nuisance mode is added inside a public reimplementation of the Union3 likelihood—not as an official, internally validated Union3/DESI systematic model. These tests are not a replacement for the official DESI+CMB+Union3 MCMC likelihood with an internally modelled systematic.

- The full-likelihood cross-check uses Planck NPIPE CamSpec (not the official DESI `plik` choice) and omits CMB lensing; the compressed cross-check represents CMB by a Gaussian prior on $\Omega_m h^2$.

- The Union3 compressed product distributes a posterior under a spline-node prior [7]; this affects absolute contour placement more than relative model comparisons.

- The low-z template component overlaps redshift ranges where peculiar velocities and survey-composition changes can introduce coherent distance distortions; the component decomposition (Section 7.5) localizes the CPL-absorbing power to $z \approx 0.10$–0.20 rather than the lowest-z peculiar-velocity zone, but does not fully exclude a low-z systematic contribution.

- These tests do not prove that DESI dynamical dark energy is false. They identify a robustness concern in the Union3 contribution.
- Analysis was conducted with AI assistance. All numerical results are reproducible from public data and scripts.

10 Recommended Falsification Test

The definitive test is to rerun the official DESI + CMB + Union3 w_0–w_a likelihood with an added low-z / mid-z redshift-dependent nuisance parameter and compare ΛCDM+nuisance, CPL, and CPL+nuisance. If CPL loses AIC support after the nuisance is included, this would indicate that the Union3 contribution to the w_0–w_a preference is strongly sensitive to this systematic-like direction, and would be consistent with a systematics-limited SN contribution. If the preference persists, the concern is resolved. The full-likelihood cross-check of Section 5 carries out this comparison in the public pipeline and finds that CPL does lose AIC support ($+8.42 \to -0.38$) once the nuisance is marginalized; an official internal-systematics treatment is the remaining step. The three-version progression (Section 6) suggests that as Union3.1 corrections are incorporated into the full joint likelihood, the CPL preference may weaken further, consistent with the weakening already observed in official Union3.1 joint analyses.

11 Conclusion

The Union3 data show sensitivity to a low-z / mid-z nuisance direction in CPL robustness tests. A single Gaussian nuisance template improves the fixed-Ω_m residuals more than twice as well as CPL ($\Delta\chi^2 = 17.12$ vs ≈ 7.52) and renders CPL AIC-disfavored once included. A compressed-prior Cobaya cross-check gives the same qualitative picture (CPL $\Delta\chi^2$ $8.29 \to 0.52$), and a full-likelihood cross-check using Planck NPIPE CamSpec, DESI DR1 BAO, and Union3 confirms it directly: the strong AIC preference for CPL (+8.42 vs ΛCDM) is removed once the nuisance is marginalized (-0.38 vs ΛCDM+nuisance), with the posterior shifting from 2.4σ to 1.6σ from ΛCDM and a recovered amplitude $a = 0.029 \pm 0.010$ matching the compressed-data value. Template-grid, random-null, and leave-one-region-out tests show that the effect is not driven by one exact functional form or by one narrow redshift interval. Across Union3, Union3.1 UNITY1.7, and Union3.1 UNITY1.8, the nuisance amplitude decreases monotonically and the CPL AIC gain becomes small, suggesting that the Union3 contribution to the DESI DR1 w_0–w_a preference is sensitive to a low-z / mid-z systematic direction that is reduced, but not removed, in Union3.1. These results motivate a full likelihood reanalysis with an explicit redshift-dependent nuisance parameter.

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